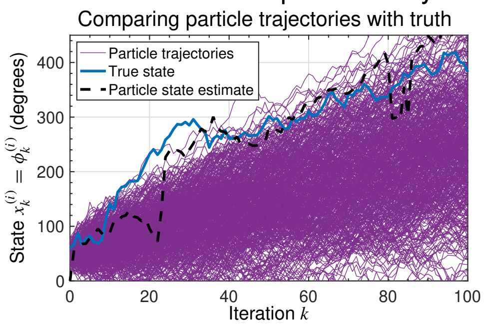
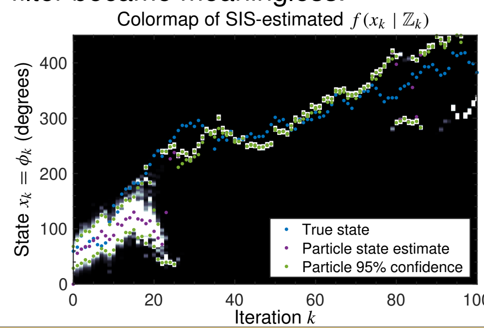
Reviewing the problem

- In the example from Lesson 4.3.3, we observed that many of the particles evolved very differently from the true state.
- The importance weights of most of the particles therefore dropped to (nearly) zero, and so the number of effective particles dropped to 1 after only about 20 iterations.
- Confidence bounds produced by the particle filter became meaningless.





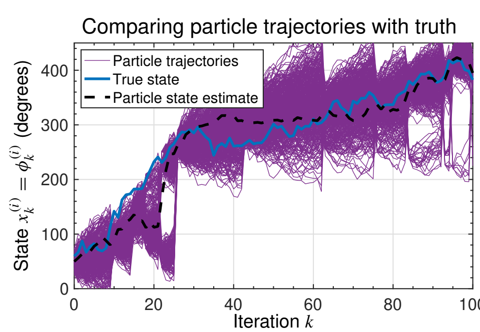
Code to resample particles

- In the prior lesson, you learned how to resample from the set of particles to improve the number of effective particles.
- Here is how we can implement this resampling method in Octave code:

```
nEff = 1/sum(wp(:,k).^2);    % compute number of effective particles
if nEff < nEffThreshold      % if lower than threshold, then resample
  xpNew = zeros(nPart,1);    % initialize new particles
  c = cumsum(wp(:,k));       % form cumulative sum of present weights
  u1 = rand(1)/nPart;        % randomly select u1~U(0,1/nPart)
  i = 1;                     % initialize particle # to replicate
  for j = 1:nPart            % find where uj falls within "c"
    uj = u1 + (j-1)/nPart;   % value to find inside of cdf "c"
    while uj > c(i) && i < nPart % search for uj inside of "c"
      i = i + 1;             % increment "i" to index the next element of "c"
    end
    xpNew(j) = xp(i,k);      % assign this new x(j) = old x(i)
  end
  xp(:,k) = xpNew;           % overwrite old particles with new particles
  wp(:,k) = ones(nPart,1)/nPart; % assign equal weights
end
```

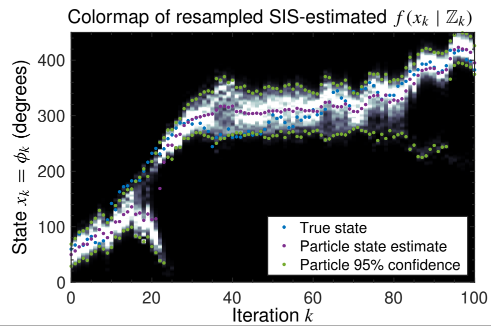

Have particle trajectories improved?

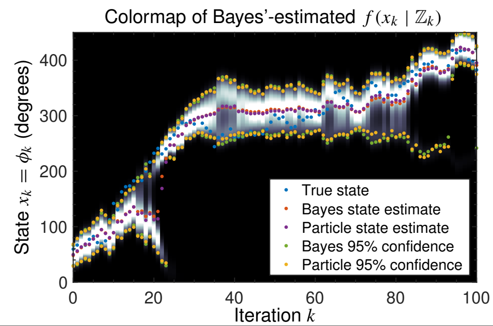
- Repeating the previous example (using exactly the same system data), with resampling threshold constraint $N_{p_e,k} \geq 100$, we obtain the results on this and the following slides.
- We notice that the particle trajectories have improved a great deal!
- Now, there are almost always many particles surrounding the true state of the system.
- So, we might assume that more particles are participating in producing the mean state estimate (confirmed later).



Has the estimated pdf improved?

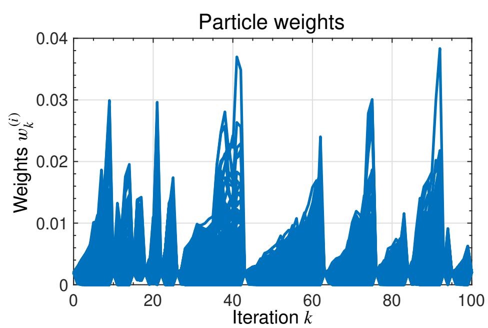
- The estimated marginal posterior from the resampled-SIS algorithm is visualized using the "bump kernel" method (left).
- Compared with the Bayes'-optimal pdf visualized on the right, we have now achieved very close agreement—large improvements in both the state estimates and confidence bounds.

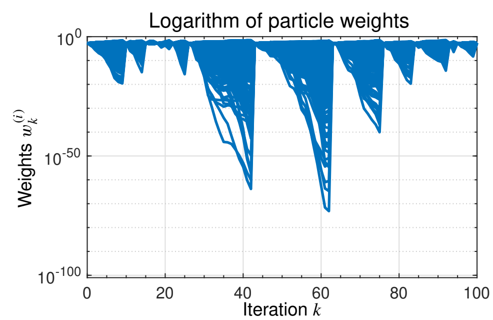




Have the particle weights improved?

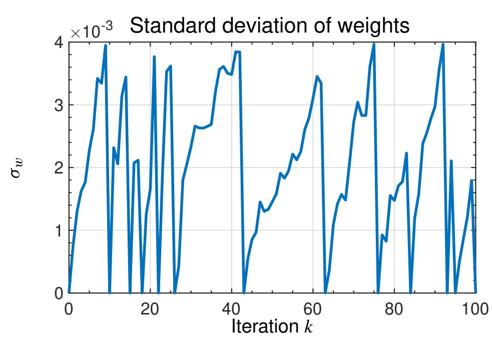
- Previously, most particle weights dropped to nearly zero; one particle weight grew to nearly one.
- When we plot the particle weights now, we find that they are much more consistent.
- There is a good variety in weights: fewer are nearly zero and none get very large.

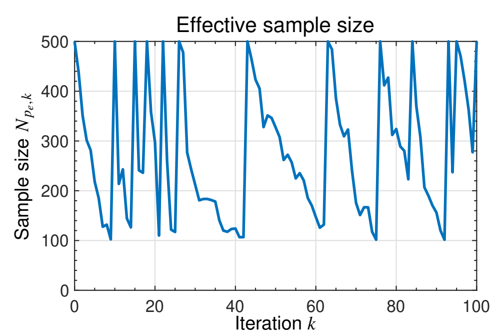




Have the weight standard deviations improved?

- Previously, the weight standard deviation grew to a limiting maximum value and the number of effective particles quickly dropped to one.
- Both the weight standard deviation and effective number of particles have improved.





Have numeric estimates improved?

- The following table lists some numeric results comparing the brute-force Bayes'-optimal method, the basic SIS method, and the SIS method with resampling.

	Root-mean-squared error of mean (rad)	Accuracy of confidence bounds
Bayes'-optimal result	0.735	84.158 %
SIS result	1.506	18.812 %
SIS with resampling result	0.835	83.168 %

- We see that the SIS-with-resampling method closely approximates the Bayes'-optimal results, but at a reduced computational demand.
- So, with resampling, particle filters become a much more effective methodology.

Summary

- This extended example has illustrated some important properties of the particle filter.
- You learned that so long as we implement a resampling method, it can produce near-optimal results, but overcomes the "curse of dimensionality" by using Monte–Carlo methods instead of brute-force Bayesian methods.
  - Resampling improves particle trajectories, the pdf estimate, and the ranges and standard deviation of particle weights.
  - It also improves numeric statistics like mean absolute error, mean squared error, and the percentage of time the confidence bounds contain the true state mean.
- You have also learned how to implement resampling in Octave; you are well prepared to write your own code to implement particle filters now!